

HACCP plan and scientific support

Ready-to-eat beef jerky

Carne Seca Jesus Canales, LLC
411 E Main St, Delta, UT 84624
(435) 406-1178

What is in this file, in print order

1. This cover.
2. Note to the reviewer.
3. HACCP plan, with the process chart, the limits chart, and the blank logs.
4. Scientific support for each critical limit.

Print every page on letter paper. Sign in ink before the packet is sent.

Sign the HACCP plan, section 8.1, Jesús Canales.

Sign the scientific support, section 6, Jesús Canales.

Leave the validation date blank until lots from this gas oven, this load, and this thickness have been reviewed.

Gas oven. Cook in a sealed bag, then dehydrate.
A laboratory test is not the release step.

Note to the reviewer

The May HACCP plan for Carne Seca Jesus Canales, LLC was returned with a request for the scientific support for the critical control points. The support had to identify the criteria used to set each critical limit for a safe ready-to-eat product.

This packet contains the HACCP plan and the scientific support.

The product is ready-to-eat beef jerky made in this plant's gas oven. The oven vent stays open, so the sealed-oven humidity option in Appendix A is not used. Lethality is a cook in a sealed moisture-impermeable bag to an internal temperature of 160°F or above, with the internal temperature between 50°F and 130°F for 6 hours or less. Dehydration starts after that cook. The dry bulb is 170°F or above, and the finished water activity is 0.85 or less, using the highest of at least six pieces.

The page for each number is in the scientific-support document. A finished-product laboratory test is not the release step. The in-plant validation date is blank until logs from this oven, this load, and this thickness have been reviewed and the plan is signed.

Jesús Canales
Carne Seca Jesus Canales, LLC
411 E Main St, Delta, UT 84624
(435) 406-1178

HACCP PLAN

Heat-treated, shelf-stable, ready-to-eat beef jerky

Carne Seca Jesus Canales, LLC

411 E Main St, Delta, UT 84624
(435) 406-1178

Process category: Heat-Treated — Shelf Stable. 9 CFR 417.2(b)(1)(vi).

Equipment: the establishment's gas oven. The oven vent stays open.

This plan replaces the returned May plan.

1. What this plan is

This plan is for the beef jerky made at this plant, in this gas oven. The meat is cooked while it is still moist, inside a sealed moisture-impermeable bag. The bag is then opened and the same oven dehydrates the meat. The finished jerky is ready to eat and shelf-stable.

A lot is released when the critical-control-point records meet the limits in Section 6 and the pre-shipment review is signed. A finished-product laboratory test is not the release step. The scientific reasons for the numbers are in the companion document, Scientific Support for the Critical Limits. That document names the hazard, the number, and the page the number comes from.

The May plan was returned because the critical limits were not tied to a stated criterion. Its cooking limit was 160°F with no come-up time, no moisture criterion, and no page citation. This plan keeps 160°F as the internal temperature and adds the two criteria Appendix A requires with that table: come-up time, and moisture maintained in a sealed bag. Drying is its own critical control point.

2. Charts

Chart 1 is the process. Chart 2 is the limits. Both are the same numbers as Section 6.

Process flow — ready-to-eat beef jerky

Carne Seca Jesus Canales, LLC · Gas oven · Delta, UT

1 Receiving

Raw beef from an approved supplier.

2 Cold storage · prerequisite

Meat at or below 41°F (5°C). Jerky field checklist.

3 Slice and marinate · prerequisite

Written formula. Meat stays at or below 41°F.

4 Seal the bag

Moisture-impermeable bag. Probe in the thickest piece.

CCP 1 Lethality · gas oven

Bag stays sealed.

Internal temperature 160°F or above.

From 50°F to 130°F in 6 hours or less.

5 Open the bag

Opened only after CCP 1 has passed.

CCP 2 Dehydration · same gas oven

Dry bulb 170°F or above during drying.

Water activity 0.85 or less.

Highest of at least six pieces.

6 Pack and label

Contains: Soy, Wheat. Net weight. Address.

7 Pre-shipment review

Records signed before the lot leaves.

If a CCP limit is missed, the lot is held and is not sold as ready-to-eat jerky.

Chart 1. Process flow.

Critical limits

Heat-treated, shelf-stable beef jerky · gas oven · no laboratory release test

Point	Limit	Where the number comes from
CCP 1 Lethality	Internal 160°F or above, thickest piece, coldest spot. Readings are not averaged.	Appendix A, Table 2. At 158°F and above the reduction is instantaneous. This plan uses 160°F.
CCP 1 Come-up	Internal temperature is between 50°F and 130°F for 6 hours or less.	Appendix A, page 24. The temperature is internal.
CCP 1 Moisture	Sealed moisture-impermeable bag until the internal temperature has reached 160°F.	Appendix A, page 31. Moisture stays in the bag, so humidity is not monitored as its own limit. A gas oven is not sealed (page 29).
CCP 2 Drying	Dry bulb 170°F or above for the whole drying stage.	Jerky model CCP 2, and the jerky checklist dry-bulb temperature. Drying starts after CCP 1 passes.
CCP 2 Water activity	0.85 or less. Highest of at least six pieces.	Preservation chapter: <i>S. aureus</i> growth minimum 0.85, and as low as 0.86 with oxygen. Checklist is 0.88.
Prerequisite Cooler	Raw meat at or below 41°F.	Jerky field checklist, cooler row. This is not a critical control point.

Chart 2. Critical limits and the criterion for each number.

3. Product description

Item	This product
Name	Beef jerky (carne seca). Ready to eat. Heat-treated. Shelf-stable. The bag may also say Traditional Beef Jerky.
Process	Whole-muscle beef, sliced, marinated, sealed in a moisture-impermeable bag, cooked in the gas oven to an internal temperature of 160°F or above, then dehydrated in the same oven to a water activity of 0.85 or less.
Equipment	Gas oven. The burner vent stays open, so the oven is not a sealed oven. Appendix A, page 29, says the sealed-oven humidity method is not used when an opening cannot be closed.
Water activity	Finished jerky: 0.85 or less.
pH	Not a control. This jerky is not fermented and is not acidified.
Cure	Uncured. No nitrite, no nitrate, no erythorbate.
Intended use	Ready to eat by the general public. No further cooking. 9 CFR 417.2(a)(2).
Package	Heat-sealed food-grade bag. The bag states the net weight of the jerky in that bag. The retail unit used on the returned plan is 3 oz (85 g).
Storage on the bag	Unopened: cool, dry, at or below 70°F (21°C). Once opened: refrigerate and consume within 3 days. Safety of the unopened jerky is the water activity of 0.85 or less, not the 70°F line.

Shelf life	The number of days is written here after this plant's own holding records support it: _____ days.
Where sold	Local retail, markets, and direct sale to household consumers.
Allergens	Soy and wheat, from Worcestershire sauce and soy sauce. Every bag says Contains: Soy, Wheat.

3.1 Ingredients — the same list as the bag

Beef, sea salt, garlic powder, onion powder, black pepper, paprika, Worcestershire sauce (vinegar, molasses, corn syrup, water, soy sauce (water, soybeans, wheat, salt), onion, garlic, cloves, tamarind, natural flavor), soy sauce (water, soybeans, wheat, salt), apple cider vinegar, liquid smoke.

Contains: Soy, Wheat.

Salt is for flavor. It does not replace the cook or the water-activity limit. No ingredient is added after the bag is opened. A batch whose weights do not match the batch sheet for this list is not cooked.

3.2 Label

- Product name includes Beef Jerky. English is required. “Carne seca” may be added.
- Ingredient statement matches Section 3.1, in descending order of predominance.
- Contains: Soy, Wheat.
- Net weight, the inspection legend and establishment number, the address 411 E Main St, Delta, UT 84624, and a handling statement consistent with a ready-to-eat product.
- The word “Traditional” is on the current artwork. Before that artwork is used under inspection, the establishment confirms whether the word is a special statement or claim that requires sketch approval under 9 CFR 412.1(c). A bag with no special claim is generically approved under 9 CFR 412.2 when every required feature is present and true.
- The safe-handling label for raw meat is not used. This jerky is ready to eat.
- The name “jerky” has a standard of identity. The moisture-to-protein ratio is 0.75:1 or less. That ratio is checked on validation lots for the name on the bag. It is not a critical limit.

Before a lot ships, the packer checks the bag against the batch sheet. A bag that fails that check is not sold.

4. Process steps

Step	What happens
1. Receiving	Raw beef enters from an approved supplier.
2. Cold storage	The meat is held at or below 41°F. This is a prerequisite.
3. Slicing	Whole muscle is sliced. Bone and foreign material are removed by looking at the meat.
4. Marination	The formula in Section 3.1 is weighed onto the batch sheet. The meat stays at or below 41°F.
5. Bag	Strips go into a moisture-impermeable bag. A probe is placed in the thickest piece. The bag is sealed so moisture stays inside, including around the probe entry.

6. CCP 1 — Lethality	The sealed bag goes into the heated gas oven. The limits in Section 6 apply. Drying does not start until this step has passed.
7. Open	The bag is opened only after the internal temperature of 160°F or above is recorded.
8. CCP 2 — Dehydration	The strips are racked in the same gas oven. Dry bulb stays at 170°F or above. The lot is dried until water activity is 0.85 or less.
9. Cool and handle	Finished jerky is handled only in the clean ready-to-eat area. Bare hands do not touch it.
10. Pack and label	The bag is checked against Section 3.2.
11. Storage	Dry storage. The safety control is the water activity already met at CCP 2.
12. Pre-shipment review	The lot's records are reviewed and signed before the lot ships.

The flow was walked in the plant on _____ by _____.

5. Hazard analysis

9 CFR 417.2(a). A hazard that is controlled by a prerequisite, and is not reasonably likely to occur because of that program, is not a critical control point. Biological hazards at the cook and at drying are reasonably likely to occur and are the two critical control points.

Step	Hazard	CCP?	Reason
1 Receiving	Biological: pathogens on raw beef	No	Approved supplier. The cook is CCP 1.
2 Cold storage	Biological: outgrowth	No	Prerequisite. Meat at or below 41°F. Jerky checklist cooler row.
3 Ingredients	Chemical: soy, wheat; nitrite	No	The formula contains soy and wheat and no nitrite. The batch sheet and the label check control the allergen statement.
4 Packaging	Biological: contamination	No	Food-grade bags, stored clean and dry.
5 Slicing	Physical: bone, metal. Biological: contamination from people	No	Visual inspection while slicing. Employee health and hygiene prerequisite.
6 Marination	Biological: outgrowth. Chemical: wrong allergen statement	No	Held at or below 41°F. Weights match Section 3.1.
7–8 CCP 1 Cooking	Biological: STEC (O157:H7, O26, O45, O103, O111, O121, O145), Salmonella, L. monocytogenes	Yes	Raw beef may carry these organisms. The bag keeps the surface moist while the internal temperature reaches 160°F.
9 CCP 2 Drying	Biological: C. perfringens and C. botulinum while the meat is still moist; S. aureus toxin; L. monocytogenes in storage	Yes	Drying at 170°F or above, then water activity 0.85 or less.
10 Handle	Biological: L. monocytogenes after the cook	No	Sanitation SOP in the ready-to-eat area. Growth in storage is prevented by CCP 2, because 0.85 is below the 0.92 growth minimum.
11 Pack	Chemical: undeclared soy or wheat	No	Every bag is checked for Contains: Soy, Wheat before the lot ships.

12 Storage	Biological: growth	No	Water activity is already 0.85 or less.
13 Returns	Biological: unknown holding	No	Opened bags are not resold.

6. Critical control points

9 CFR 417.2(c). Every limit in a critical control point is required. The criterion for each number is named here and is set out page by page in the scientific-support document. 9 CFR 417.3: a miss identifies the cause, brings the point back under control, stops it from happening again, and keeps injurious product out of commerce.

6.1 CCP 1 — Lethality in the gas oven

The bag is sealed and the meat is still moist. Drying does not start until every CCP 1 limit has passed. The probe is in the thickest piece, in the coldest part of the load. That cold spot is confirmed during initial validation. Readings are not averaged.

Critical limit	Criterion	Monitoring, each lot
Internal temperature 160°F or above.	Appendix A, December 2021, Table 2, footnote 5. The required reduction is achieved instantly when the internal temperature reaches 158°F or above. This plan uses 160°F, which is above that line. The temperature is the minimum in the meat, not the oven air.	Time the probed piece reaches 160°F, the temperature read, the probe identity, and the place in the oven.
The internal temperature is between 50°F and 130°F for 6 hours or less.	Appendix A, page 24. Come-up time is an internal temperature. The meat enters from a cooler at or below 41°F, so it enters below 50°F.	Clock time the probed piece reaches 50°F and clock time it reaches 130°F. The difference is written in hours and minutes.
The moisture-impermeable bag stays sealed until the internal temperature is 160°F or above.	Appendix A, page 31. Cooking in a sealed moisture-impermeable bag keeps moisture around the product, so relative humidity is not monitored as its own limit. Page 29: this gas oven cannot use the sealed-oven method.	The seal is checked when the bag enters the oven and again at the 160°F reading. The time the bag is opened is written. It is not opened before that reading.

Who monitors: the person running the oven, named on the log.

If any CCP 1 limit is missed: the lot is stopped and Jesús Canales is told. The lot is held. It is not opened for dehydration and it does not ship. The cause is written. The oven or the bag practice is corrected before the next lot. Product that missed the cook is not sold as ready-to-eat jerky. A laboratory test is not the release step.

The cook may continue inside the same sealed bag only when the piece can still reach 160°F and the time between 50°F and 130°F is still 6 hours or less. Appendix A states that a come-up clock does not start over when the first cook did not reach a lethal temperature. If that time is already over 6 hours, the lot is not released under this plan.

6.2 CCP 2 — Dehydration

This step starts only after CCP 1 has passed. The bag is opened and the strips are racked so the gas oven can dry them.

Critical limit	Criterion	Monitoring, each lot
Dry-bulb temperature 170°F or above during drying, read before the jerky is taken out.	The jerky field checklist prints a dry bulb above 170°F (77°C). The FSIS beef-jerky model, CCP 2, uses an oven temperature of 170°F or above so <i>Clostridium perfringens</i> and <i>Clostridium botulinum</i> do not grow while water activity is still above 0.93. That model cites the FSIS Stabilization Guideline, Revised Appendix B, December 2021. The checklist's wet-bulb line is not a limit in this plan. Appendix A, page 27, says a wet bulb of 125–130°F for one hour is not, by itself, the humidity support for Appendix A.	Dry-bulb reading during drying, thermometer identity, and the time.
Finished water activity 0.85 or less. At least six pieces, from different places in the lot, including a thick piece. The highest reading is the lot result. Every piece checked is 0.85 or less.	Principles of Preservation of Shelf-Stable Dried Meat Products: the aerobic growth minimum for <i>Staphylococcus aureus</i> is 0.85, and it can grow as low as 0.86 when oxygen is present. The same table gives <i>Escherichia coli</i> O157:H7 a minimum of 0.95, <i>Salmonella</i> 0.94, <i>Listeria monocytogenes</i> 0.92, and <i>Clostridium perfringens</i> and proteolytic <i>Clostridium botulinum</i> 0.93. The jerky checklist prints water activity below 0.88. A result of 0.85 or less meets that sheet. The jerky section of the preservation chapter says safety is judged by water activity, not by the moisture-to-protein ratio.	After drying, before packing. Calibrated water-activity meter. All six readings are written. The highest is circled.

Who monitors: the person running the oven, named on the log.

If the dry bulb is below 170°F, or if any of the six readings is above 0.85: the lot is not packed. Jesús Canales is told. If CCP 1 passed, the lot may be dried longer at 170°F or above and measured again. If it still fails, it does not ship. It is not wetted and cooked again under this plan.

7. Prerequisite programs

These programs are part of the HACCP system. They are not critical control points. A failure is corrected under the program and recorded. If a failure creates an unforeseen hazard, 9 CFR 417.3(b) applies: the product is held, the cause is found, injurious product does not ship, and a trained person reassesses whether the hazard belongs in the plan.

- Cold storage. Raw beef for this jerky stays at or below 41°F (5°C), measured in the meat. The jerky field checklist prints this cooler limit. The FSIS model's example prerequisite is below 45°F. This plan uses 41°F.
- Employee health. The establishment uses the FDA Employee Health Policy Tool (2022 Food Code, part 2-201; tool dated August 23, 2023) for restriction and exclusion of ill employees. A person who is ill does not handle meat or jerky. Hands are washed. Bare hands do not touch finished jerky. This tool does not set a critical limit.

- Sanitation SOP, 9 CFR 416. After the bag is opened, jerky is handled only in the clean area. The sanitation SOP names Alternative 2 of 9 CFR 430.4: the finished water activity of 0.85 or less is the antimicrobial process that prevents growth of *Listeria monocytogenes*, together with sanitation in the ready-to-eat area. The food-contact-surface testing frequency required for that alternative is written in the sanitation SOP before a lot ships.
- Allergens. The batch sheet lists soy and wheat. The pack-out check confirms the bag says Contains: Soy, Wheat.
- Thermometers and the water-activity meter are calibrated on the schedule in Section 8. An instrument that fails calibration is removed, and the lots checked with it since the last good calibration are reviewed.

The curing-and-smoking field checklist is not a prerequisite for this jerky and is not support for these limits. This product is not brine-cured, is not dry-cured, and is not hung to dry before a smoke schedule. That checklist was read so the wrong process would not be copied into this plan.

8. Verification and records

- Initial validation, 9 CFR 417.4(a)(1). The scientific support is the companion document and the pages it cites. The in-plant part is repeated lots on this gas oven, this load, and this thickness, showing that CCP 1 and CCP 2 were met and showing the cold spot. The validation date below stays blank until those logs have been reviewed.
- Ongoing verification. Each production week, a person other than the monitor watches one CCP check when product is being made. Thermometers are checked in ice water at 32°F. The water-activity meter is checked with the manufacturer's standards. Records are reviewed before shipment.
- Reassessment, 9 CFR 417.4(a)(3). At least once a year, and when the oven, the bag, the formula, the thickness, or the load changes. The responsible official signs the reassessment.
- Pre-shipment review, 9 CFR 417.5(c). Before the lot ships, a person reviews that lot's records, writes that every critical limit was met or that the corrective action is finished, and signs and dates the review. Where practical, that person did not produce the record. The lot does not ship until this review says it may ship.
- Records, 9 CFR 417.5. Each record shows the plant, the date, the product, the lot, the actual number measured, the initials of the monitor, and the review. Shelf-stable product records are kept for at least two years.
- Training, 9 CFR 417.7. At least one person has completed training in the seven HACCP principles for meat and poultry. Name: _____ Date of training: _____.

8.1 Signature

9 CFR 417.2(d). Signing accepts this plan for this product and this gas oven.

Responsible establishment official, Jesús Canales: _____

Date: _____

Initial validation completed (blank until the in-plant lots are reviewed):

Annual reassessment: _____

9. Forms

9.1 CCP 1 log — lethality

Lot _____ Date _____ Oven _____ Probe _____

Bag sealed on entry: yes / no. Probe in the thickest piece: yes / no. Cold-spot location: _____

Reading	Clock time	Internal °F	Initials
Internal temperature reaches 50°F			
Internal temperature reaches 130°F			
Come-up (130 minus 50), hours and minutes. Limit: 6 hours or less.			
Internal temperature reaches 160°F or above			
Bag still sealed at that reading, then time opened			

Limit met: yes / no. If no, lot held and Form 9.4 is completed. Reviewer _____ Date _____

9.2 CCP 2 log — dehydration

Lot _____ Date _____ Dry-bulb thermometer _____ Water-activity meter _____

CCP 1 passed before the bag was opened: yes / no.

Dry bulb during drying: _____ °F. Limit: 170°F or above. Time read: _____

Piece	Place in the lot	Water activity	Initials
1			
2			
3			
4			
5			
6 (include a thick piece)			

Highest reading: _____. Limit: 0.85 or less. Limit met: yes / no. Reviewer _____ Date _____

9.3 Cooler log — prerequisite, not a CCP

Date _____ Time _____ Product temperature _____ °F Limit: 41°F or less Initials _____

9.4 Corrective action

Lot _____ CCP _____ Date _____ Time _____

What was measured: _____

Limit that was missed: _____

Cause: _____

Action that brought the point back under control: _____

Action that keeps it from happening again: _____

Disposition. The lot is held. It is not sold as ready-to-eat jerky unless every limit is met on the records. A laboratory test is not the release step.

Jesús Canales: _____

9.5 Pre-shipment review

Lot _____ Date _____

- CCP 1: 160°F internal, come-up 6 hours or less, bag sealed until that reading.
- CCP 2: dry bulb 170°F or above, highest of six water-activity readings 0.85 or less.
- Bag says Contains: Soy, Wheat, and the ingredients match the batch sheet.
- Any corrective action is finished. No injurious product is in this shipment.

Reviewer signature and date: _____

10. Documents kept with this plan

- Scientific Support for the Critical Limits, the companion to this plan.
- FSIS Cooking Guideline for Meat and Poultry Products (Revised Appendix A), December 2021, FSIS-GD-2021-14.
- Principles of Preservation of Shelf-Stable Dried Meat Products, October 31, 2011.
- HACCP Field Verification Checklist, Jerky (fully cooked, shelf-stable, ready-to-eat).
- FDA Employee Health Policy Tool, August 23, 2023, for the employee-health prerequisite.
- FSIS HACCP Model for Ready-to-Eat, Heat-Treated, Shelf-Stable Beef Jerky, 2021-0004, CCP 2, for the 170°F drying temperature.
- FSIS Stabilization Guideline (Revised Appendix B), December 2021, as cited by that model.

The curing-and-smoking checklist is on file as a document that was reviewed and not used. The returned May plan is on file as the plan this one replaces.

SCIENTIFIC SUPPORT**Criteria used to set each critical limit**

Ready-to-eat, heat-treated, shelf-stable beef jerky

Carne Seca Jesus Canales, LLC · 411 E Main St, Delta, UT 84624

This is the support requested for the critical control points. For each critical limit it names the hazard, the number, the criterion used to choose that number, and the page where that criterion is written. It is written so the reviewer can see why each limit is the limit for this ready-to-eat product.

The product is whole-muscle beef jerky cooked in this plant's gas oven and then dehydrated in that same oven. The oven vent stays open. The jerky is not fermented, is not acidified, and contains no nitrite. Nothing is added after the heat step. pH is not a critical limit. Soy and wheat are in the formula and are declared on the bag. They are controlled by the batch sheet and the label check. They are not a critical control point.

A lot is released on the process records. This support does not rely on a finished-product laboratory test, and it does not cite a challenge study of this jerky. 9 CFR 417.4(a)(1) still has an in-plant part. The validation date on the HACCP plan stays blank until logs from this oven, this load, and this thickness have been reviewed.

1. The process the limits belong to

Stage	What the plant does	Why it is in this order
CCP 1, lethality	Strips are sealed in a moisture-impermeable bag and heated in the gas oven until the internal temperature is 160°F or above, with a come-up of 6 hours or less between 50°F and 130°F.	Appendix A says humidity has to be applied during lethality, before drying (page 26). The sealed bag is the way this oven keeps the surface moist (page 31).
CCP 2, dehydration	The bag is opened only after CCP 1 passes. The gas oven then dries the strips at a dry bulb of 170°F or above until water activity is 0.85 or less.	Drying is what makes the jerky shelf-stable. It is not the lethality step.

2. CCP 1 — lethality**2.1 Internal temperature 160°F or above**

- Hazard. Salmonella and Shiga-toxin producing Escherichia coli, including O157:H7, on raw beef, including on the surface of a sliced strip. Listeria monocytogenes is included in the cook because the product is ready to eat.
- Critical limit. Internal temperature 160°F or above, in the thickest piece, in the coldest part of the load. Readings are not averaged. The oven-air temperature is not the limit.
- Criterion. FSIS Cooking Guideline for Meat and Poultry Products (Revised Appendix A), December 2021, FSIS-GD-2021-14, Table 2, page 35, "Time-Temperature Combinations for Meat Products to Achieve Lethality." Footnote 5 on that page: the required log reductions are achieved

instantly (0 seconds) when the internal temperature of a cooked meat product reaches 158°F or above. The row at 145°F on the same table is 4 minutes for both the 6.5-log column and the 7-log column. This plan does not use the 145°F row. It uses 160°F, which is above the 158°F instantaneous line, so the dwell time at 160°F is zero on that table.

- The same table states that relative humidity and come-up time are also critical operating parameters when the table is used. Those two parameters are the next two limits. They are not optional.
- The preservation chapter, page 167, says processors must validate a 5-log reduction of *E. coli* O157:H7 for products that contain beef, and that the product must be heated to achieve it. The heat step in this plan is the Table 2 endpoint above.

2.2 Come-up time, 6 hours or less

- Hazard. *Staphylococcus aureus* growth and heat-stable enterotoxin if the meat stays too long between 50°F and 130°F during heating.
- Critical limit. The time the internal temperature is between 50°F and 130°F is 6 hours or less.
- Criterion. Appendix A, page 24: “Come-Up-Time Option: Total time product temperature is between 50 and 130°F is 6 hours or less.” The same page says those temperatures are internal temperatures. Footnote 7 of Table 2, page 35, repeats the 6-hour limit.
- The meat is held at or below 41°F before it is cooked, so it enters the oven below 50°F. The clock starts when the probed piece reaches 50°F and stops when it reaches 130°F.

2.3 Sealed moisture-impermeable bag

- Hazard. *Salmonella* surviving on a surface that dries before the lethal internal temperature is reached. Drying the surface early makes the organism harder to kill.
- Critical limit. The strip stays in a sealed moisture-impermeable bag until the internal temperature is 160°F or above. The bag is opened only after that reading is recorded. Dehydration starts after that.
- Criterion. Appendix A, page 31, “Situations when Humidity is Not Needed.” Moisture is inherently maintained when the product is cooked in a sealed, moisture-impermeable bag. Establishments that match that situation do not monitor relative humidity as its own critical operating parameter. The bag is that situation for this gas oven.
- Appendix A, page 26: humidity is applied during the lethality treatment, before drying. Opening the bag only after 160°F puts dehydration after lethality.
- Appendix A, page 29: if an oven has an opening that cannot be closed, the sealed-oven method is not used. This is a gas oven. The burner vent stays open. The plan therefore does not use Humidity Option 2.
- Appendix A, page 27: a wet-bulb temperature of 125–130°F and 27–32 percent relative humidity for one hour, taken from the jerky guideline, is not adequate on its own to support the Appendix A humidity options. This plan does not use that pair as the humidity limit. The jerky checklist

prints a wet bulb of 125–142°F next to its heat-lethality line. That line is not a critical limit in this plan, for the reason on page 27.

- The preservation chapter, Jerky Products section, page 168, says the jerky guidelines call for a humidity step at the beginning of the process. The sealed-bag cook is that step. Dehydration follows it.

3. CCP 2 — dehydration

3.1 Dry bulb 170°F or above

- Hazard. Outgrowth of *Clostridium perfringens* and *Clostridium botulinum* while water activity is still high enough for growth, during the drying stage.
- Critical limit. The dry-bulb temperature of the gas oven is 170°F or above throughout drying, read before the jerky is removed.
- Criterion. The jerky field checklist, heat-lethality row, prints a dry bulb above 77°C (170°F). This plan applies that dry-bulb temperature to the whole drying stage. The FSIS HACCP Model for Ready-to-Eat, Heat-Treated, Shelf-Stable Beef Jerky, 2021-0004, CCP 2, sets the oven at 170°F or above during drying so those sporeformers do not grow while water activity is still above 0.93, and it cites the FSIS Stabilization Guideline, Revised Appendix B, December 2021, for that temperature. Lethality itself is CCP 1, not this dry-bulb reading.

3.2 Water activity 0.85 or less

- Hazard. *Staphylococcus aureus* growth and toxin on a jerky that is stored in air. *Listeria monocytogenes* growth during storage. Growth of the vegetative pathogens named in the preservation table if water activity stays above their minima.
- Critical limit. Finished water activity 0.85 or less. At least six pieces from different places in the lot, including a thick piece. The highest reading is the lot result.
- Criterion. Principles of Preservation of Shelf-Stable Dried Meat Products, October 31, 2011, growth-minimum table (chapter pages 157–158): *Staphylococcus aureus*, aerobic, 0.85. The same chapter, page 160: *Staphylococcus aureus* can grow as low as 0.86 water activity when oxygen is present. A finished result of 0.85 or less is at or below that aerobic minimum and below 0.86. The same table: *Escherichia coli* O157:H7, 0.95; *Salmonella*, 0.94; *Listeria monocytogenes*, 0.92; *Clostridium perfringens*, 0.93; proteolytic *Clostridium botulinum*, 0.93. Finished jerky at 0.85 or less is below each of those minima.
- The same chapter, page 160: dried hams, coppa, and beef jerky generally have water activity less than 0.88. The jerky field checklist, drying row, prints water activity below 0.88. The limit in this plan is 0.85 or less, which meets the checklist.
- The same chapter, Jerky Products section, page 168: safety and shelf stability of jerky are judged by water activity, not by the moisture-to-protein ratio, and the older practice of relying on a moisture-to-protein ratio of 0.75 or below is not the safety indicator. Page 160 states that

moisture-to-protein ratios are labeling standards and are not necessarily indicative of microbial safety. This plan checks 0.75:1 on validation lots for the product name. It is not a critical limit.

- The chapter also states that if pathogens are still viable, the product is adulterated. Water activity stops growth. It is not the kill step. The kill step is CCP 1.

4. Limits that are not critical control points

Item	Number used	Why it is not a CCP
Cooler	Raw meat at or below 41°F (5°C).	Jerky field checklist, cooler-storage row. It keeps the meat below the start of the come-up range. Outgrowth before the cook is controlled by this prerequisite.
Employee health	Restriction and exclusion under the 2022 FDA Food Code, part 2-201, using the Employee Health Policy Tool dated August 23, 2023.	People who are ill do not handle the product. The tool does not identify a process critical limit.
Allergens	Contains: Soy, Wheat, on every bag. Formula in the plan, Section 3.1.	Controlled by the batch sheet and the pack-out check.
Moisture-to-protein ratio	0.75:1 or less, on validation lots.	Standard of identity for the name jerky. Preservation chapter, page 160 and page 168. Not a safety limit.
Curing and smoking checklist	Not used.	That checklist is for brine, dry cure, a salinometer, and hanging meat before smoke. This jerky is not that process. Using it would attach the wrong criteria to these critical limits.

5. What the returned May plan did not show

The returned plan set CCP 1 at 160°F and CCP 2 at a water activity of 0.85 or less. It did not identify the Table 2 footnote that makes 158°F instantaneous, it did not set the 6-hour come-up, and it did not state how moisture is maintained in this gas oven. Section 2 of this document is that identification. The water-activity number is kept and is now tied to the preservation-chapter table and to the checklist value it also satisfies.

6. Signature

I am providing this document as the scientific support for the critical limits in the HACCP plan for this ready-to-eat beef jerky.

Responsible establishment official, Jesús Canales: _____

Date: _____